

Entropy formula:-

$$H(Y) = - \sum_{i=1}^m P_i \log(P_i)$$

$$* \text{Info}(D) = \sum_{i=1}^m P_i \log_2(P_i)$$

$$* \text{Info}_A(D) = \sum_{j=1}^V \frac{|D_j|}{|D|} \times \text{info}(D_j)$$

$$\text{Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$$

→ class P :- buys - computer = "Yes"

→ class N :- buys - computer = "No"

Age	class		I(P, N)
	P	N	
<30	2	3	0.9710
31-40	4	0	0
>40	3	2	0.9710
Total	9	5	0.9401

$$I(2,3) = - \sum P_i \log(P_i)$$

$$= - \frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right)$$

$$= - \frac{2}{5} (-1.322) - \frac{3}{5} (-0.737)$$

$$= 0.5288 + 0.4422$$

$$= 0.9710$$

$$I(4,0) = -\frac{4}{4} \log_2(1) = 0$$

$$= 0$$

$$\begin{aligned} I(3,2) &= -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \\ &= -\frac{3}{5} (-0.737) - \frac{2}{5} (-0.322) \\ &= 0.4422 + 0.5288 \\ &= 0.9710 \end{aligned}$$

$$\begin{aligned} I(9,5) &= -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\ &= -\frac{9}{14} (-0.6374) - \frac{5}{14} (-1.4854) \\ &= 0.4097 + 0.5304 \\ &= 0.9401 \end{aligned}$$

$$\begin{aligned} \text{Info}_{\text{age}}(0) &= \frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971 \\ &= 0.3467 + 0 + 0.3467 \\ &= 0.6934 \end{aligned}$$

$$\begin{aligned} \text{Gain}(\text{Age}) &= 0.9401 - 0.6934 \\ &= 0.2467 \end{aligned}$$

income	class		$I(p, n)$
	p	n	
high	2	2	1
medium	4	2	0.918
low	3	1	0.811
	9	5	0.9401

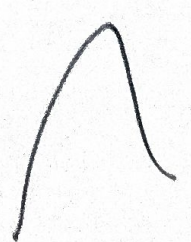
$$\begin{aligned} I(2,2) &= -\sum p_i \log_2(p_i) \\ &= -\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right) \\ &= -\frac{1}{2} (-1) - \frac{1}{2} (-1) \\ &= 1 \end{aligned}$$

$$\begin{aligned}
 I(4,2) &= -\frac{4}{6} \log_2(4/6) - \frac{2}{6} \log_2(2/6) \\
 &= -\frac{4}{6}(-0.585) - \frac{2}{6}(-1.585) \\
 &= 0.39 + 0.528 \\
 &= 0.918
 \end{aligned}$$

$$\begin{aligned}
 I(3,1) &= -\frac{3}{4} \log_2(3/4) - \frac{1}{4} \log_2(1/4) \\
 &= -\frac{3}{4}(-0.415) - \frac{1}{4}(-2) \\
 &= 0.311 + 0.5 \\
 &= 0.811
 \end{aligned}$$

$$\begin{aligned}
 I(9,5) &= -\frac{9}{14} \log_2(9/14) - \frac{5}{14} \log_2(5/14) \\
 &= -\frac{9}{14}(-0.6374) - \frac{5}{14}(-1.4854) \\
 &= 0.9401
 \end{aligned}$$

$$\begin{aligned}
 I_{\text{so}_{\text{age}}}(D) &= \frac{4}{14} \times 1 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 0.811 \\
 &= 0.285 + 0.393 + 0.231 \\
 &= \cancel{1.484} \quad 0.909
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain}(\text{Age}) &= 0.9401 - 0.909 \\
 &= 0.0311
 \end{aligned}$$


$$\text{Gain}(\text{Student}) = 0.151$$

$$\text{Gain}(\text{Credit-rating}) = 0.048$$

$$\text{Gain}(\text{income}) = 0.0391$$

	gain	ratio	index
age	0.246	0.156	0.443
income	0.03	1.019	
student	0.151		
credit	0.48		

$$\text{gain ratio}(A) = \text{gain}(A) / \text{splitinfo}(A)$$

$$\text{splitinfo}_A(D) = \sum_{j=1}^n \frac{|D_j|}{|D|} \times \log_2 \left(\frac{|D_j|}{|D|} \right)$$

$$\begin{aligned} \text{splitinfo}_{\text{age}} &= -\frac{5}{14} \log_2 \frac{5}{14} - \frac{4}{14} \log_2 \frac{4}{14} - \frac{5}{14} \log_2 \frac{5}{14} \\ &= 0.5097 + 0.5304 + 0.5304 + 0.5163 \\ &= 1.5771 \approx 1.58 \end{aligned}$$

$$\text{splitinfo}_{(\text{income})} = \frac{0.246}{1.58} = 0.156$$

$$\begin{aligned} \text{splitinfo}_{(\text{income})} &= -\frac{4}{14} \log_2 \frac{4}{14} - \frac{6}{14} \log_2 \frac{6}{14} - \frac{4}{14} \log_2 \frac{4}{14} \\ &= 0.5163 + 0.5163 + 0.5238 \\ &= 1.5564 \end{aligned}$$

$$\text{gain ratio} = 1.019$$

Gini index:-

$$\text{Gini}(D) = 1 - \sum_{j=1}^n p_j^2$$

$$\text{Gini}_A(D) = \frac{|D_1|}{|D|} \text{gini}(D_1) + \frac{|D_2|}{|D|} \text{gini}(D_2)$$

$$\begin{aligned}
 \text{Gini}_D &= 1 - \left[\left(\frac{9}{14} \right)^2 + \left(\frac{5}{14} \right)^2 \right] \\
 &= 1 - [0.413 + 0.127] \\
 &= 1 - 0.54 \\
 &= 0.460
 \end{aligned}$$

Partition the data set wrt income

$$D_1 = \{ \text{low, medium} \}_{10}, \quad D_2 = \{ \text{High} \}_{4}$$

$$\begin{aligned}
 \text{Gini}_A(D) &= \frac{10}{14} \left[1 - \left(\frac{7}{10} \right)^2 - \left(\frac{3}{10} \right)^2 \right] + \frac{4}{14} \left[1 - \left(\frac{2}{4} \right)^2 - \left(\frac{2}{4} \right)^2 \right] \\
 &= 0.714 [1 - 0.49 - 0.09] + 0.285 [1 - 0.25 - 0.25] \\
 &= 0.714 [0.42] + 0.285 [1 - 0.5] \\
 &= 0.30 + 0.1425 \\
 &= 0.44
 \end{aligned}$$

Take low high together & medium separate

$$D_1 = \{ \text{low, high} \}_{8}$$

$$D_2 = \{ \text{medium} \}_{6}$$

$$\begin{aligned}
 \text{Gini}_A(D) &= \frac{8}{14} \left[1 - \left(\frac{5}{8} \right)^2 - \left(\frac{3}{8} \right)^2 \right] + \frac{6}{14} \left[1 - \left(\frac{4}{6} \right)^2 - \left(\frac{2}{6} \right)^2 \right] \\
 &= 0.571 [1 - 0.39 - 0.14] + 0.428 [1 - 0.43 - 0.10] \\
 &= 0.571 [0.47] + 0.428 [0.47] \\
 &= 0.267 + 0.2011 \\
 &= 0.4681
 \end{aligned}$$

Take high medium together & low separate

$$\text{Gini}_A(D_1) = \{ \text{high, medium} \}_{10}$$

$$D_2 = \{ \text{low} \}_{4}$$

$$Gini_A(D) = \frac{10}{14} \left[1 - \left(\frac{6}{10} \right)^2 - \left(\frac{4}{10} \right)^2 \right] + \frac{4}{14} \left[1 - \left(\frac{3}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right]$$

$$= \frac{10}{14} [0.48] + \frac{4}{14} [0]$$

$$= 0.3430$$

LM	H	0.443
LH	M	0.458
MH	L	0.450

a Confusion matrix:

Actual class / Predicted class	cancer = yes	cancer = no	Total
cancer = yes	90 (TP)	210 (FN)	300 (P)
cancer = no	140 (FP)	9560 (TN)	9700 (N)
Total	230 (P')	9770 (N')	10000 (all)

$$\rightarrow \text{accuracy} = \frac{90 + 9560}{10000} = \frac{9650}{10000} \times 100 = 96.5$$

$$\rightarrow \text{Error rate} = 100 - 96.5 = 3.5 \quad (\%) = 1 - 0.965 = 0.035 \times 100 = 3.5$$

$$\rightarrow \text{sensitivity} = \frac{90}{300} = 0.3 \times 100 = 30$$

$$\rightarrow \text{Specificity} = \frac{9560}{9700} = 0.985 \times 100 = 98.5$$

$$\rightarrow \text{Precision} = \frac{90}{90 + 140} = \frac{90}{230} \times 100 = 39.1$$

$$\rightarrow \text{recall} = \frac{90}{90 + 210} = \frac{90}{300} = 0.3 \times 100 = 30$$

$$\rightarrow F = \frac{2 \times 39.1 \times 30}{39.1 + 30} = \frac{2346}{69.1} = 33.95 \%$$

1. A shepherd boy gets bored herding the town's flock. To have some fun he cries out, "WOLF!" To have some insight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

A)

actual \ predicted		
	WOLF = YES	WOLF = NO
WOLF = YES	TP	FN
WOLF = NO	FP	TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 30 of the 70 no cat images. Identify TP, FP, FN, TN and calculate the precision & recall & F1 score.

A)

	Predicted cat	Predicted No cat	
Actual cat	25	5	30
Actual No cat	20	50	70
	45	55	100

$$\rightarrow \text{Precision} = \frac{TP}{TP+FP} = \frac{25}{25+20} = \frac{25}{45} = 0.556$$

$$= 55.6\%$$

$$\rightarrow \text{Recall} = \frac{TP}{TP+FN}$$

$$= \frac{25}{25+5} = \frac{25}{30} = 0.8333$$

$$= 83.33\%$$

$$\rightarrow F_1 \text{ score} = \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}}$$

$$= \frac{2(0.556)(0.8333)}{0.556 + 0.8333}$$

$$= 0.667$$

$$F_1 \text{ score} = 66.67\%$$

③ ~~Assume~~ An email service provider wants to classify emails as Spam or Not Spam using features:

- contains "offer" (Yes/No)
- contains "Free" (Yes/No)
- Email Length (Short/Long)

Sample Dataset

offer Free Length class

Yes Yes Short class

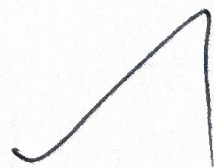
Yes No Long Not Spam

No Yes Short Spam

No No Long Not Spam

Yes Yes Long Spam

No No Short Not Spam



A) 1) Total spam = 6

spam = 3, Not spam = 3

$$P(\text{spam}) = \frac{3}{6} = 0.5$$

$$P(\text{Not spam}) = \frac{3}{6} = 0.5$$

2) Conditional probabilities:-

For spam:-

$$P(\text{offer} = \text{yes} / \text{spam}) = \frac{2}{3}$$

$$P(\text{free} = \text{yes} / \text{spam}) = \frac{3}{3} = 1$$

$$P(\text{Length} = \text{short} / \text{spam}) = \frac{2}{3}$$

For Not spam:-

$$P(\text{offer} = \text{yes} / \text{Not spam}) = \frac{1}{3}$$

$$P(\text{free} = \text{yes} / \text{Not spam}) = \frac{0}{3} = 0$$

$$P(\text{Length} = \text{short} / \text{Not spam}) = \frac{1}{3}$$

3) Classify a new email:-

(offer = yes, free = No, Length = short)

For spam:-

$$P(\text{spam} / x) \propto P(\text{spam}) \times P(\text{yes} / \text{spam}) \times P(\text{No} / \text{spam}) \times P(\text{short} / \text{spam})$$

$$= 0.5 \times \frac{2}{3} \times 0 \times \frac{2}{3} = 0$$

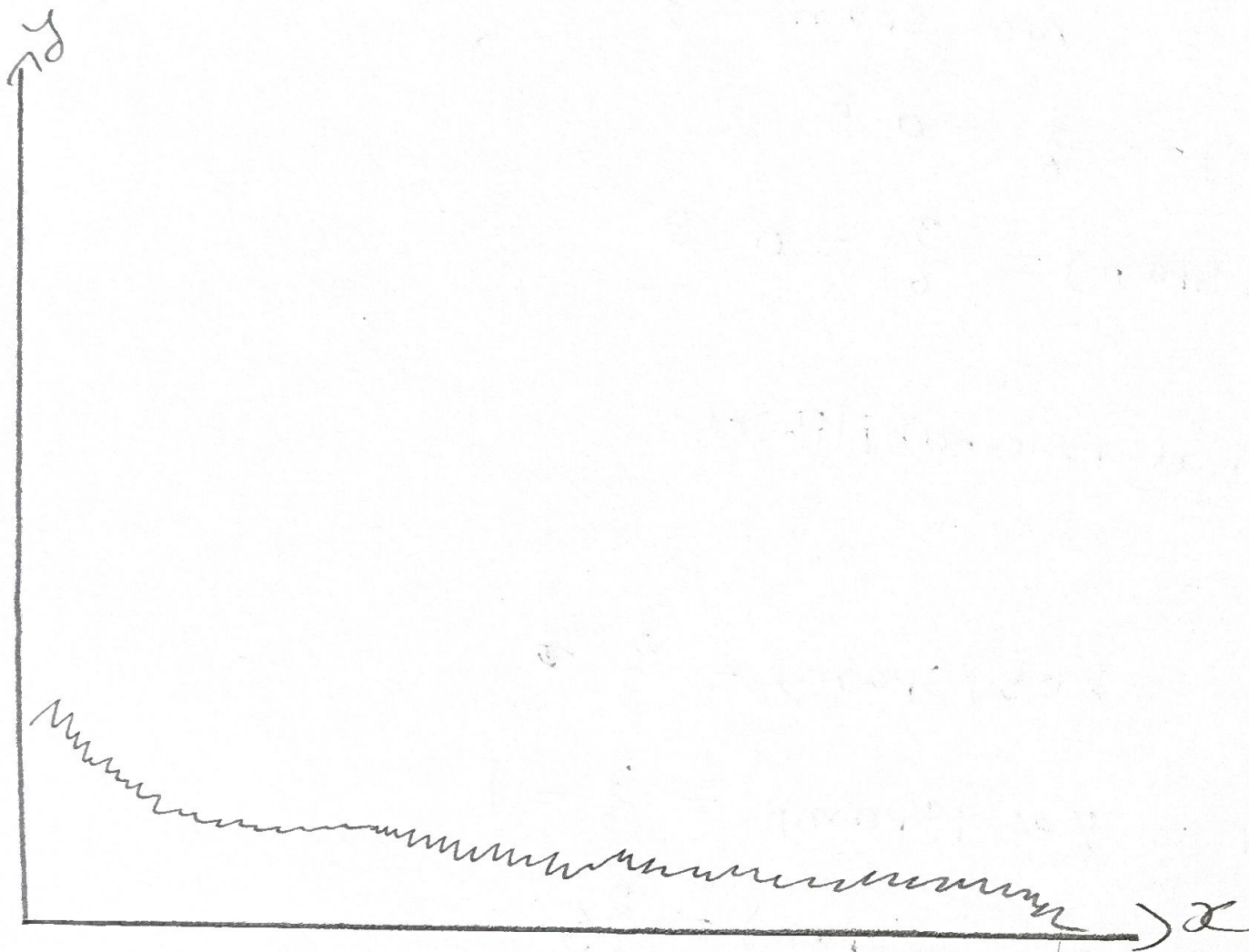
For Not spam

$$P(\text{Not spam} / x) \propto 0.5 \times \frac{1}{3} \times 1 \times \frac{1}{3}$$

$$= \frac{1}{18} = 0.0556$$

Prediction = Not spam

④ compare results using WAKB



✓
Chopain
a/8/16